Zomato Restaurant Analytics Dashboard

1. Project Overview

The Zomato Restaurant Analytics Dashboard is an end-to-end data analytics project developed using Python, SQLite, SQL, and Power BI. The project aims to analyze restaurant data and uncover meaningful insights related to customer ratings, pricing, cuisines, restaurant types, online ordering services, table booking facilities, and restaurant popularity.

The project follows a complete analytics workflow involving data cleaning, exploratory data analysis (EDA), database creation using SQLite, SQL-based business analysis, and dashboard development using Power BI.

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2. Objectives

The primary objectives of this project are:

- Analyze restaurant ratings and customer preferences.
- Identify the most popular restaurant types and cuisines.
- Study the impact of online ordering and table booking on ratings.
- Discover high-density restaurant locations.
- Analyze restaurant pricing patterns.
- Build an interactive dashboard for business decision-making.

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3. Technologies Used

Programming and Analysis

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook

Database

- SQLite
- SQL

Business Intelligence

- Power BI

Version Control

- Git
- GitHub

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4. Dataset Description

Dataset Information

- Original Records: 7,105
- Records After Cleaning: 6,984

Features Used

- Restaurant Name
- Restaurant Type
- Rate (Out of 5)
- Number of Ratings
- Average Cost (Two People)
- Online Order
- Table Booking
- Cuisines Type
- Area
- Local Address

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5. Data Cleaning and Preprocessing

Several preprocessing steps were performed to improve data quality and ensure accurate analysis.

Cleaning Steps

- Removed unnecessary columns:
    - Unnamed: 0
    - Unnamed: 0.1
- Identified and handled missing values.
- Removed rows containing missing values in critical columns.
- Checked for duplicate records.
- Standardized the dataset for analysis.

Final Dataset

- Total Records: 6,984
- Total Features: 10

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6. Feature Engineering

A new metric called **Popularity Score** was created to measure restaurant influence and popularity.

Formula

Popularity Score = Rating × Number of Ratings

This metric helps identify restaurants that have both high ratings and a large number of customer reviews.

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7. Exploratory Data Analysis (EDA)

The following analyses were performed using Python, Pandas, Matplotlib, and Seaborn:

Restaurant Analysis

- Top Restaurant Types
- Top Rated Restaurant Categories
- Restaurant Popularity Analysis

Customer Service Analysis

- Online Ordering Distribution
- Table Booking Distribution
- Impact of Online Ordering on Ratings
- Impact of Table Booking on Ratings

Location Analysis

- Top Areas by Number of Restaurants

Cuisine Analysis

- Most Popular Cuisines

Rating Analysis

- Rating Distribution
- Average Ratings by Category

Cost Analysis

- Cost Distribution

- Cost vs Rating Relationship
- Average Cost by Restaurant Type

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8. SQLite Database Integration

To demonstrate database management and SQL analytics skills, the cleaned dataset was stored in a SQLite database.

Database Workflow

Cleaned Dataset (CSV)
→ SQLite Database
→ SQL Queries
→ Query Results
→ Python Visualizations

The SQLite database contains a table named:

- zomato

The database was created using Python's sqlite3 library.

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9. SQL Analysis

SQL queries were used to perform business-focused analysis on the restaurant dataset.

SQL Business Questions

1. What is the total number of restaurants?
2. What is the average restaurant rating?
3. What is the average cost for two people?
4. Which areas contain the highest number of restaurants?
5. What percentage of restaurants offer online ordering?
6. What percentage of restaurants offer table booking?
7. How does online ordering affect restaurant ratings?
8. How does table booking affect restaurant ratings?
9. Which restaurant types have the highest average ratings?
10. Which restaurant types have the highest average cost?

SQL Tools Used

- SQLite
- SQL Queries
- Pandas read_sql_query()

10. Power BI Dashboard Development

An interactive dashboard was created using Power BI to visualize business insights.

KPI Cards

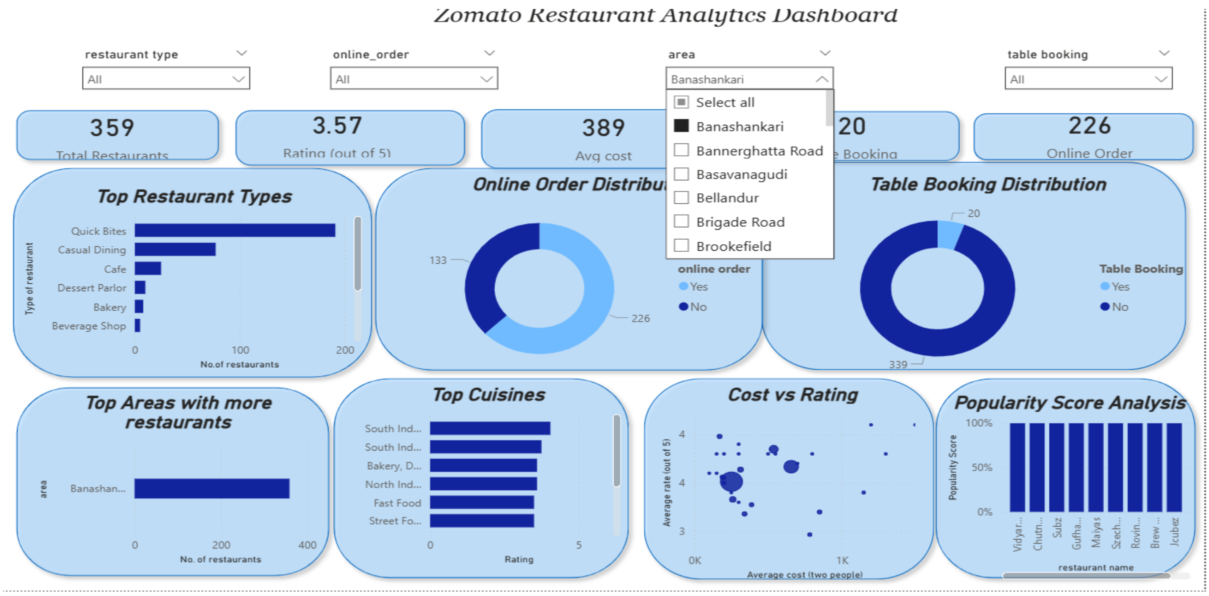
- Total Restaurants
- Average Rating
- Average Cost for Two People
- Online Order Restaurants
- Table Booking Restaurants

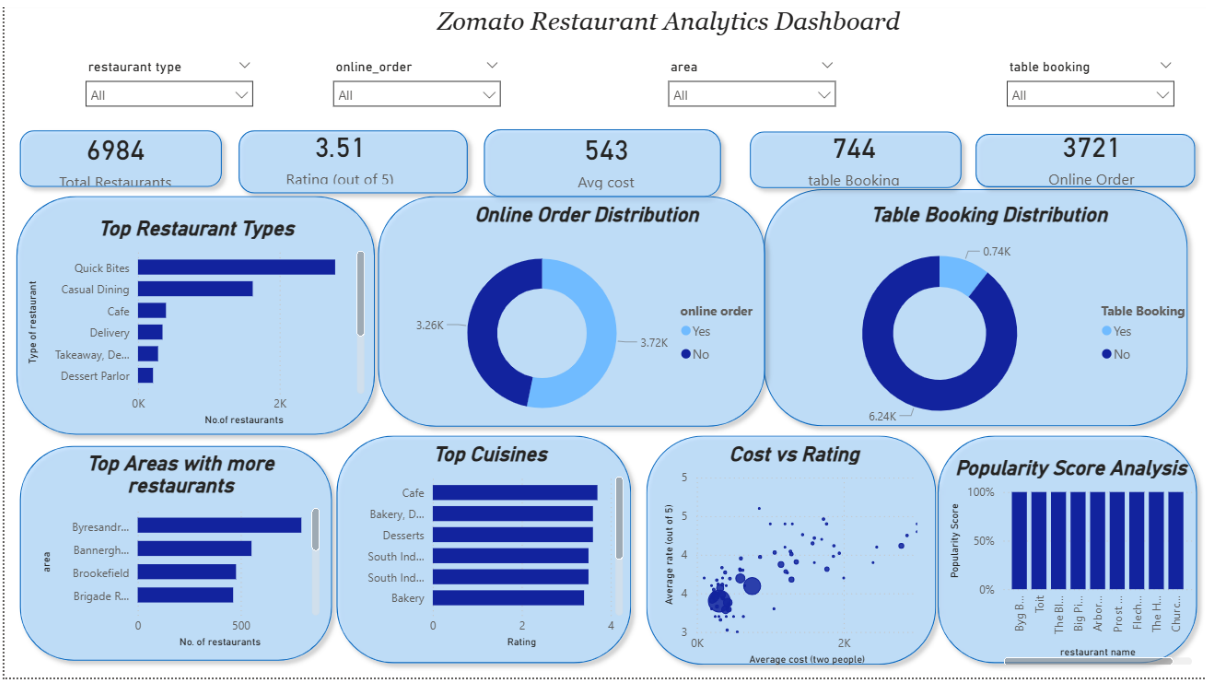
Visualizations

- Top Restaurant Types
- Online Order Distribution
- Table Booking Distribution
- Top Areas
- Top Cuisines
- Cost vs Rating Analysis
- Popularity Score Analysis
- Cuisine-wise Average Rating

Interactive Filters

- Area
- Restaurant Type
- Online Order
- Table Booking





11. Key Findings

- Quick Bites is the most common restaurant type.
- More than 53% of restaurants provide online ordering services.
- Restaurants offering table booking have significantly higher average ratings.
- The average restaurant rating is approximately 3.51 out of 5.

- The average cost for two people is approximately ₹543.
- Byresandra–Tavarekere–Madiwala has the highest restaurant concentration.
- Highly rated restaurant categories generally attract larger customer engagement.
- Popularity Score effectively identifies restaurants with strong customer impact.

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12. Project Structure

Zomato-Analytics-Project

- Dataset
- Notebook
- Database
- SQL_Queries
- PowerBI
- Reports
- Screenshots
- README.md
- requirements.txt
- .gitignore

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13. Conclusion

The Zomato Restaurant Analytics Dashboard successfully demonstrates an end-to-end analytics workflow by integrating Python, SQLite, SQL, and Power BI.

The project transforms raw restaurant data into meaningful business insights through data cleaning, exploratory analysis, SQL-based querying, visualization, and dashboard reporting. The findings help understand customer behavior, restaurant performance, pricing strategies, and service preferences, enabling data-driven decision-making.

This project also demonstrates practical skills in data preprocessing, SQL analysis, database management, data visualization, dashboard development, and business intelligence reporting.